Week3 : Queues, Deques, and Priority Queues Objectives

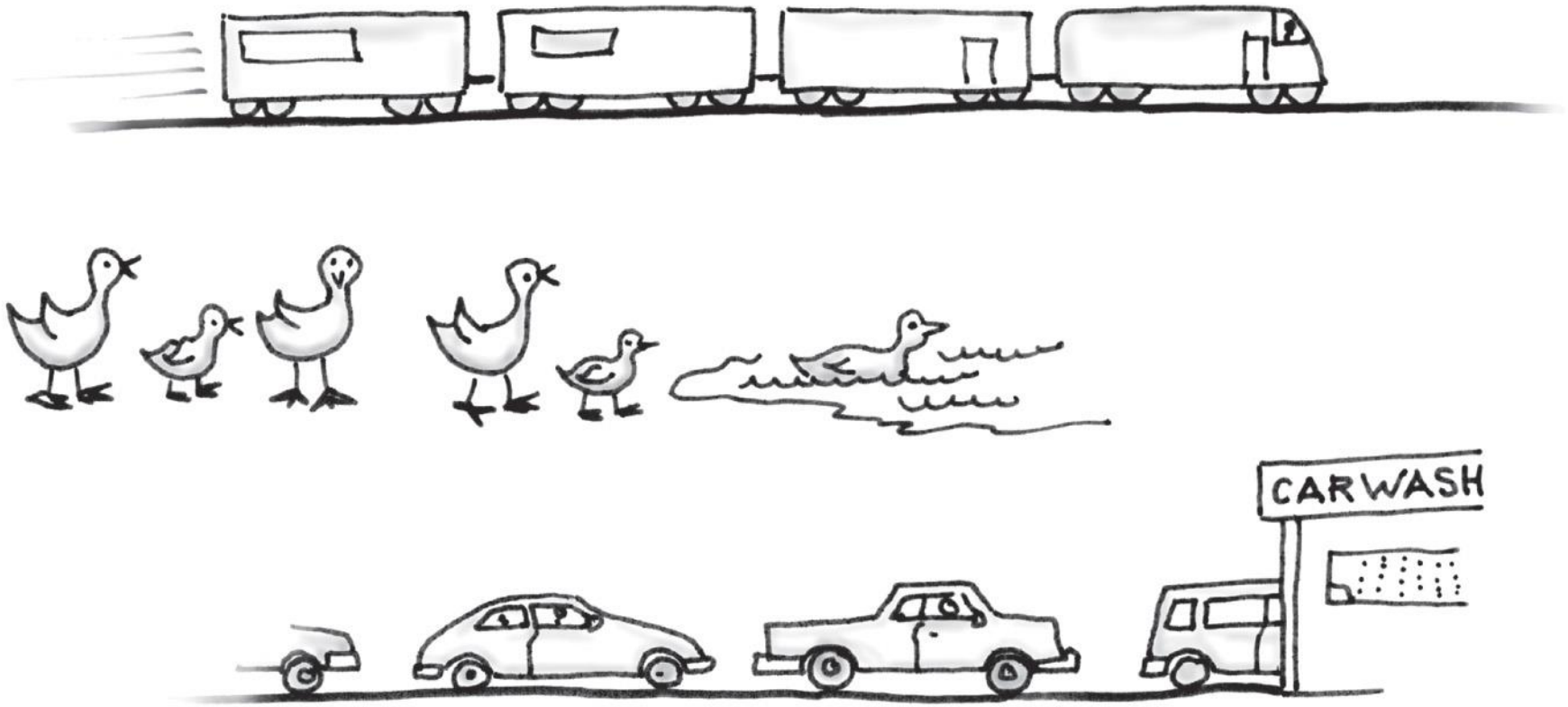
Upon completion of this unit, students will be able to understand and explain the following sub topics :

- Queues, Deques, and Priority Queues & Queue, Deque, and Priority Queue Implementations

The ADT Queue

- A queue is another name for a waiting line
- Used within operating systems and to simulate real-world events
  - Come into play whenever processes or events must wait
- Entries organized first-in, first-out

The ADT Queue



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The ADT Queue

- Terminology
  - Item added first, or earliest, is at the front of the queue
  - Item added most recently is at the back of the queue
- Additions to a software queue must occur at its back
- Client can look at or remove only the entry at the front of the queue

The ADT Queue

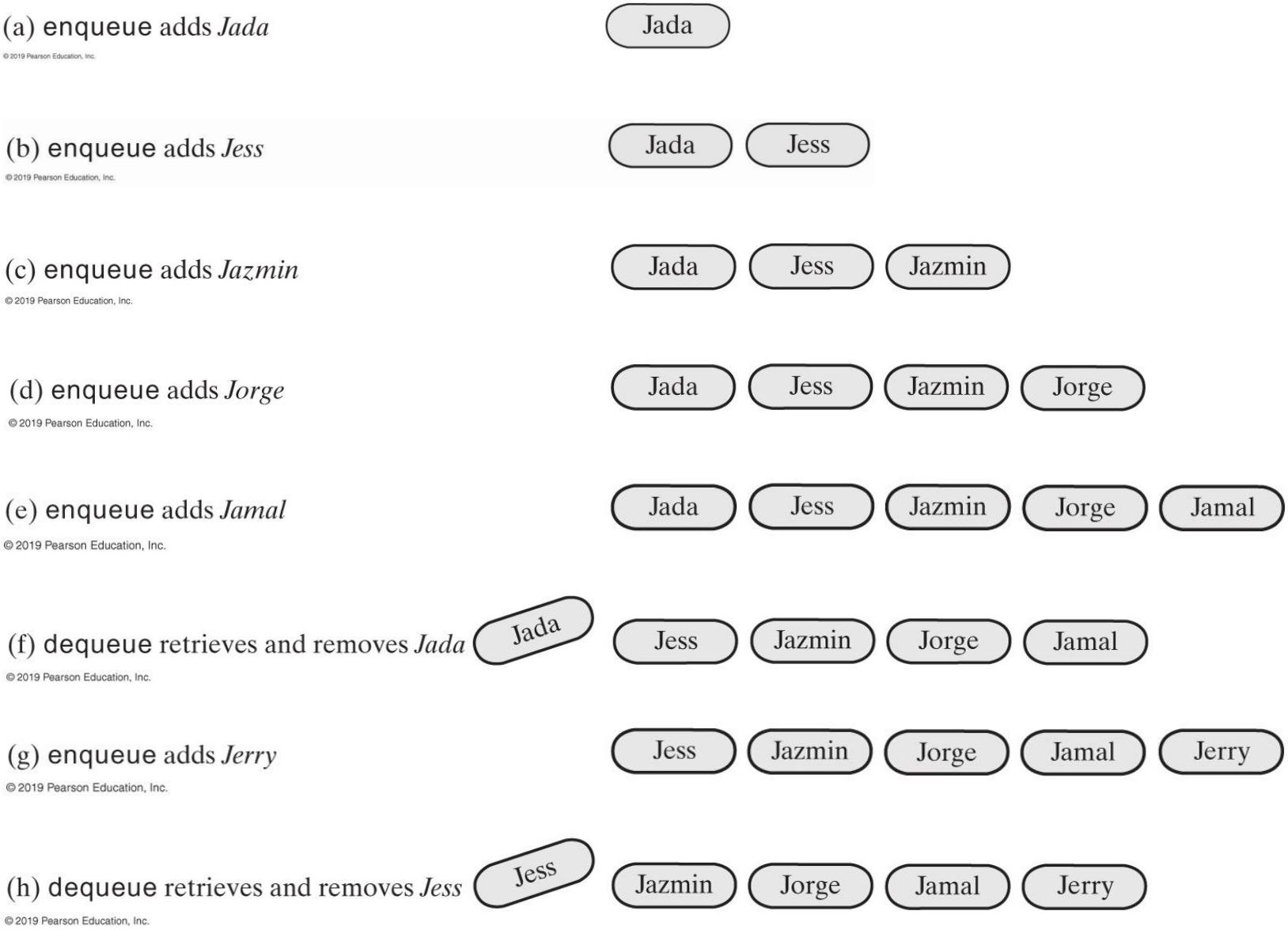
- Data
  - A collection of objects in chronological order and having the same data type

enqueue(newEntry)	+enqueue(newEntry: integer): void	Task: Adds a new entry to the back of the queue. Input: newEntry is the new entry. Output: None.
dequeue()	+dequeue(): T	Task: Removes and returns the entry at the front of the queue. Input: None. Output: Returns the queue's front entry. Throws an exception if the queue is empty before the operation.
getFront()	+getFront(): T	Task: Retrieves the queue's front entry without changing the queue in any way. Input: None. Output: Returns the queue's front entry. Throws an exception if the queue is empty.
isEmpty()	+isEmpty(): boolean	Task: Detects whether the queue is empty. Input: None. Output: Returns true if the queue is empty.
clear()	+clear(): void	Task: Removes all entries from the queue. Input: None. Output: None.

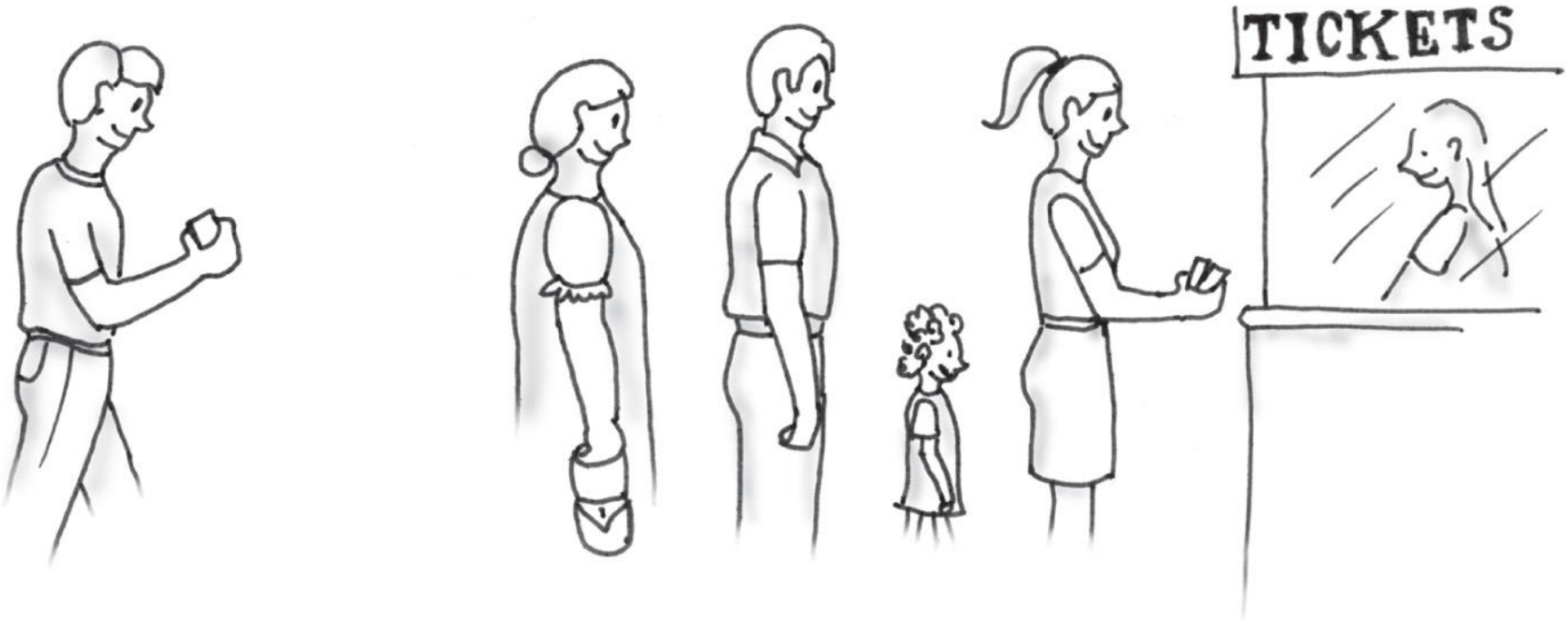
The ADT Queue

```
/** An interface for the ADT queue.  */
public interface QueueInterface<T>
{
  /** Adds a new entry to the back of this queue.
      @param newEntry  An object to be added. */
  public void enqueue(T newEntry);

  /** Removes and returns the entry at the front of this queue.
      @return  The object at the front of the queue.
      @throws  EmptyQueueException if the queue is empty before the operation. */
  public T dequeue();

  /**  Retrieves the entry at the front of this queue.
      @return  The object at the front of the queue.
      @throws  EmptyQueueException if the queue is empty. */
  public T getFront();

  /** Detects whether this queue is empty.
      @return  True if the queue is empty, or false otherwise. */
  public boolean isEmpty();

  /** Removes all entries from this queue. */
  public void clear();
} // end QueueInterface
```

FIGURE 7-2 The effect of operations on a queue of strings



Simulating a Waiting Line



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Simulating a Waiting Line

- Algorithm for simulate

```
Algorithm simulate(duration, arrivalProbability, maxTransactionTime)
transactionTimeLeft = 0
for (clock = 0; clock < duration; clock++)
{
        if (a new customer arrives)
        {
                numberOfArrivals++
                transactionTime = a random time that does not exceed maxTransactionTime
                nextArrival = a new customer containing clock, transactionTime, and
                                a customer number that is numberOfArrivals
                line.enqueue(nextArrival)
        }
        if (transactionTimeLeft > 0) // If present customer is still being served
                transactionTimeLeft−−
        else if (!line.isEmpty())
        {
                nextCustomer = line.dequeue()
                transactionTimeLeft = nextCustomer.getTransactionTime() − 1
                timeWaited = clock − nextCustomer.getArrivalTime()
                 totalTimeWaited = totalTimeWaited  +  timeWaited numberServed++
        }
}
```

Simulating a Waiting Line

```
/** Simulates a waiting line.  */
public class WaitLine
{
   private QueueInterface<Customer> line;
   private int numberOfArrivals;
   private int numberServed;
   private int totalTimeWaited;

   public WaitLine()
   {
      line = new LinkedQueue<>();
      reset();
   } // end default constructor

   /** Initializes the simulation. */
   public final void reset()
   {
      line.clear();
      numberOfArrivals = 0;
      numberServed = 0;
      totalTimeWaited = 0;
   } // end reset

   public void simulate(int duration, double arrivalProbability,
                        int maxTransactionTime)
   {   < implementation on next slide > }

   public void displayResults()
   {  < implementation on next slide >  }
 } // end WaitLine
```

- LISTING 7-2 The class WaitLine

Simulating a Waiting Line

```
public void simulate(int duration, double arrivalProbability, int maxTransactionTime)
{
  int transactionTimeLeft = 0;

  for (int clock = 0; clock < duration; clock++)
  {
    if (Math.random() < arrivalProbability)
    {
      numberOfArrivals++;
      int transactionTime = (int)(Math.random() * maxTransactionTime + 1);
      Customer nextArrival = new Customer(clock, transactionTime,
                                          numberOfArrivals);
      line.enqueue(nextArrival);
      System.out.println("Customer " + numberOfArrivals +
                         " enters line at time " + clock +
                         ". Transaction time is " + transactionTime);
    } // end if

    if (transactionTimeLeft > 0)
      transactionTimeLeft--;
    else if (!line.isEmpty())
    {
      Customer nextCustomer = line.dequeue();
      transactionTimeLeft = nextCustomer.getTransactionTime() - 1;
      int timeWaited = clock - nextCustomer.getArrivalTime();
      totalTimeWaited = totalTimeWaited + timeWaited;
      numberServed++;
      System.out.println("Customer " + nextCustomer.getCustomerNumber() +
                         " begins service at time " + clock +
                         ". Time waited is " + timeWaited);
    } // end if
  } // end for
} // end simulate
```

Simulating a Waiting Line

```
/** Displays summary results of the simulation. */
 public void displayResults()
 {
    System.out.println();
    System.out.println("Number served = " + numberServed);
    System.out.println("Total time waited = " + totalTimeWaited);
    double averageTimeWaited = ((double)totalTimeWaited) / numberServed;
    System.out.println("Average time waited = " + averageTimeWaited);
    int leftInLine = numberOfArrivals - numberServed;
    System.out.println("Number left in line = " + leftInLine);
 } // end displayResults
```

Computing the Capital Gain in a Sale of Stock (Part 1)

```
public class StockLedger
{
  private QueueInterface<StockPurchase> ledger;

  public StockLedger()
  {
    ledger = new LinkedQueue<>();
  } // end default constructor

  /** Records a stock purchase in this ledger.
     @param sharesBought   The number of shares purchased.
     @param pricePerShare  The price per share. */
  public void buy(int sharesBought, double pricePerShare)
  {
    while (sharesBought > 0)
    {
      StockPurchase purchase = new StockPurchase(pricePerShare);
      ledger.enqueue(purchase);
      sharesBought--;
    } // end while
  } // end buy
```

Computing the Capital Gain in a Sale of Stock (Part 2)

```
/** Removes from this ledger any shares that were sold
    and computes the capital gain or loss.
    @param sharesSold     The number of shares sold.
    @param pricePerShare  The price per share.
    @return  The capital gain (loss). */
 public double sell(int sharesSold, double pricePerShare)
 {
   double saleAmount = sharesSold * pricePerShare;
   double totalCost = 0;

   while (sharesSold > 0)
   {
     StockPurchase share = ledger.dequeue();
     double shareCost = share.getCostPerShare();
     totalCost = totalCost + shareCost;
     sharesSold--;
   } // end while

   return saleAmount - totalCost; // Gain or loss
 } // end sell
} // end StockLedger
```

Java Class Library: The Interface `Queue`

- Methods provided
  - `add`
  - `offer`
  - `remove`
  - `poll`
  - `element`
  - `peek`
  - `isEmpty`
  - `size`

The ADT `Deque`

- A double ended queue
- `Deque` pronounced “deck”
- Has both queue-like operations and stack-like operations

The ADT Deque

- FIGURE 7-10 An instance d of a `deque`

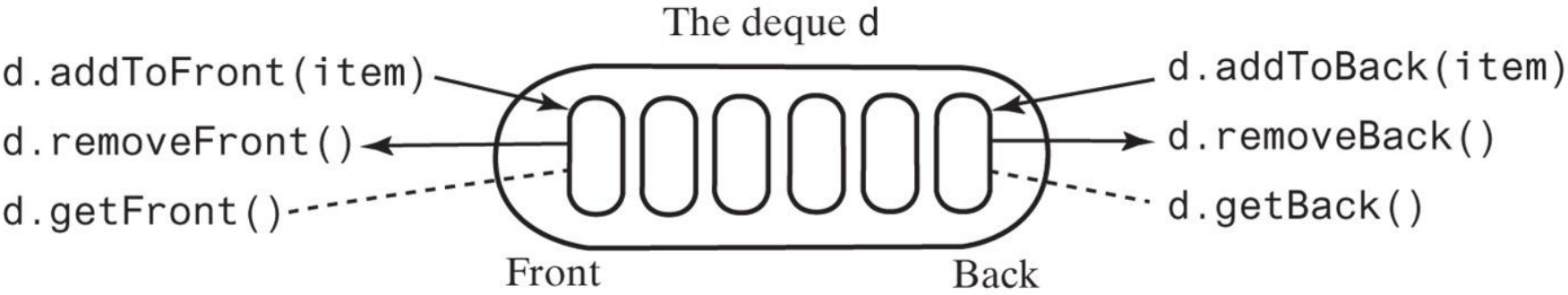

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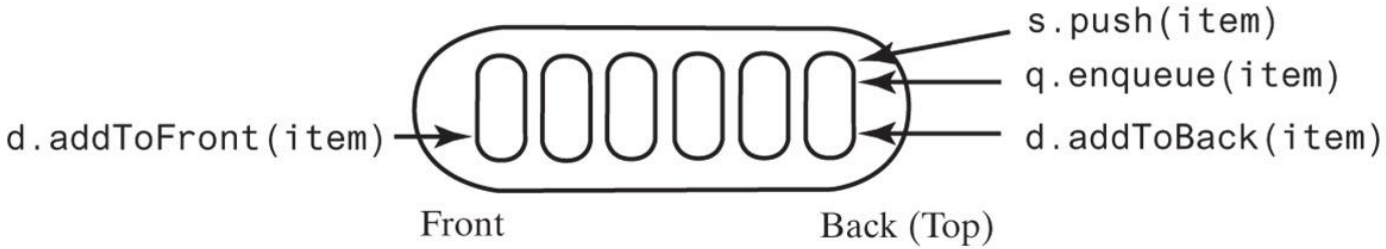
FIGURE 7-11 A comparison of operations for a stack s, a queue q, and

The stack s, queue q, or deque d

(a) Add



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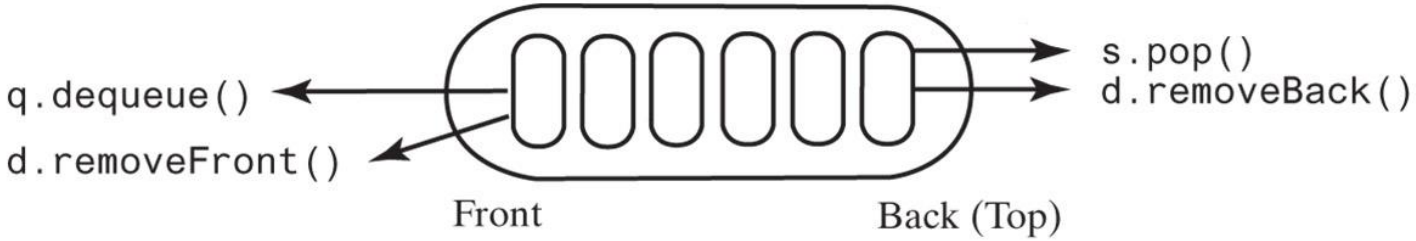
(b) Remove



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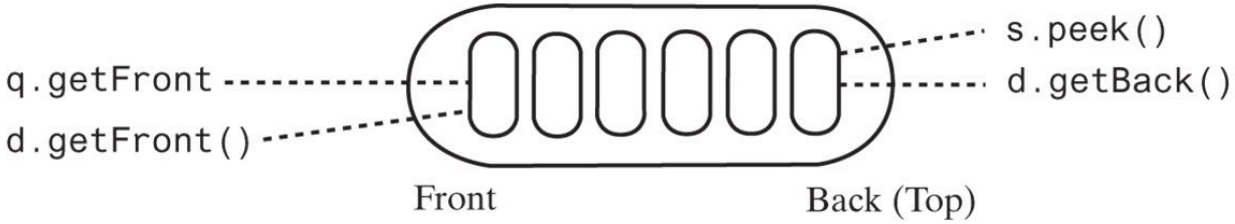
(c) Retrieve



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The ADT Deque

```
/** An interface for the ADT deque. */
public interface DequeInterface<T>
{
   /** Adds a new entry to the front/back of this deque.
      @param newEntry  An object to be added. */
   public void addToFront(T newEntry);
   public void addToBack(T newEntry);

   /** Removes and returns the front/back entry of this deque.
      @return  The object at the front/back of the deque.
      @throws  EmptyQueueException if the deque is empty before the
               operation. */
   public T removeFront();
   public T removeBack();

   /** Retrieves the front/back entry of this deque.
      @return  The object at the front/back of the deque.
      @throws  EmptyQueueException if the deque is empty. */
   public T getFront();
   public T getBack();

   /** Detects whether this deque is empty.
      @return  True if the deque is empty, or false otherwise. */
   public boolean isEmpty();

   /*  Removes all entries from this deque. */
   public void clear();
} // end DequeInterface
```

Computing the Capital Gain in a Sale of Stock

```
public double sell(int sharesSold, double pricePerShare)
{
   double saleAmount = sharesSold * pricePerShare;
   double totalCost = 0;

   while (sharesSold > 0)
   {
      StockPurchase transaction = ledger.removeFront();
      double shareCost = transaction.getCostPerShare();
      int numberOfShares = transaction.getNumberOfShares();

      if (numberOfShares > sharesSold)
      {
         totalCost = totalCost + sharesSold * shareCost;
         int numberToPutBack = numberOfShares - sharesSold;
         StockPurchase leftOver = new StockPurchase(numberToPutBack, shareCost);
         ledger.addToFront(leftOver); // Return leftover shares
         // Note: Loop will exit since sharesSold will be <= 0 later
      }
      else
         totalCost = totalCost + numberOfShares * shareCost;

      sharesSold = sharesSold - numberOfShares;
   } // end while

   return saleAmount - totalCost; // Gain or loss
} // end sell
```

- Method buy creates an instance of `StockPurchase` and places it at the back of the deque

Java Class Library: The Interface `Deque`

- Methods provided
  - `addFirst, offerFirst`
  - `addLast, offerLast`
  - `removeFirst, pollFirst`
  - `removeLast, pollLast`
  - `getFirst, peekFirst`
  - `getLast, peekLast`
  - `isEmpty, clear, size`
  - `push, pop`

Java Class Library: The Class `ArrayDeque`

- Implements the interface **`Deque`**
- Constructors provided
  - **`ArrayDeque()`**
  - **`ArrayDeque(int initialCapacity)`**

ADT Priority Queue

- Consider how a hospital assigns a priority to each patient that overrides time at which patient arrived.
- ADT priority queue organizes objects according to their priorities
- Definition of “priority” depends on nature of the items in the queue

ADT Priority Queue

```
/** An interface for the ADT priority queue. */
public interface PriorityQueueInterface<T extends Comparable<? super T>>
{
   /** Adds a new entry to this priority queue.
      @param newEntry  An object to be added. */
   public void add(T newEntry);

   /** Removes and returns the entry having the highest priority.
      @return  Either the object having the highest priority or, if the
               priority queue is empty before the operation, null. */
   public T remove();

   /** Retrieves the entry having the highest priority.
      @return  Either the object having the highest priority or, if the
               priority queue is empty, null. */
   public T peek();

   /** Detects whether this priority queue is empty.
      @return  True if the priority queue is empty, or false otherwise. */
   public boolean isEmpty();

   /** Gets the size of this priority queue.
      @return  The number of entries currently in the priority queue. */
   public int getSize();

   /** Removes all entries from this priority queue. */
   public void clear();
} // end PriorityQueueInterface
```